

Soru $S_1 = \frac{1}{2}$ $S_2 = \frac{1}{2}$ bağımsız ve bağlanmış Clebsch-Gordan katsayılarını bul

$$S_1 = \frac{1}{2} \quad m_1 = \frac{1}{2} \quad -\frac{1}{2} \quad J_{\max} = \frac{S_1 + S_2}{1} \quad J_{\min} = |S_1 - S_2| = 0$$

$$S_2 = \frac{1}{2} \quad m_2 = \frac{1}{2} \quad -\frac{1}{2} \quad J = 1, 0$$

ii) $\Rightarrow J = 1$ dursa $m = \{1, 0, -1\}$ $m = m_1 + m_2$

iii) $J = 0$ " $m = \{0\}$

$J \downarrow \quad m \downarrow \quad m_1 + m_2 = 1 \quad m_1 \quad m_2$

i) $J = 1 \quad m = 1 \Rightarrow |1, 1\rangle = | \frac{1}{2}, \frac{1}{2} \rangle$

$\leftarrow J = 1 \quad m = 0 \Rightarrow |1, 0\rangle = P_1 | \frac{1}{2}, -\frac{1}{2} \rangle + P_2 | -\frac{1}{2}, \frac{1}{2} \rangle$

$J = 1 \quad m = -1 \Rightarrow |1, -1\rangle = | -\frac{1}{2}, -\frac{1}{2} \rangle$

$\leftarrow J = 0 \quad m = 0 \Rightarrow |0, 0\rangle = P_3 | \frac{1}{2}, -\frac{1}{2} \rangle + P_4 | -\frac{1}{2}, \frac{1}{2} \rangle$

$|P_1|^2 + |P_2|^2 = 1 \quad |P_3|^2 + |P_4|^2 = 1$

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$\Rightarrow |J, m\rangle \Rightarrow |1, 1\rangle \Rightarrow J_- = J_{1-} + J_{2-} \Rightarrow$

$\Rightarrow J_{1-} | \frac{1}{2}, \frac{1}{2} \rangle + J_{2-} | \frac{1}{2}, \frac{1}{2} \rangle \Rightarrow \hbar \sqrt{2} |1, 0\rangle = \hbar \sqrt{\frac{1}{2}(\frac{3}{2} - \frac{1}{2} - \frac{1}{2})} | \frac{1}{2}, \frac{1}{2} \rangle + \hbar \sqrt{\frac{1}{2}(\frac{1}{2} + 1 - \frac{1}{2} - 1)} | -\frac{1}{2}, \frac{1}{2} \rangle$

$\Rightarrow \sqrt{2} |1, 0\rangle = | -\frac{1}{2}, \frac{1}{2} \rangle + | \frac{1}{2}, -\frac{1}{2} \rangle \Rightarrow P_1 = P_2 = \frac{\sqrt{2}}{2} = \frac{1}{\sqrt{2}}$

